BERN ECONOMIC DEVELOPMENT AGENCY CHALLENGE

Research & Expertise Navigator – Berner

Translating layperson search into academic reality across Bern's innovation ecosystem.

DEVELOPED BY
Building Warriors

POWERED BY
Apertus 70B & Swiss AI Stack

EVENT
HackApertus Liebefeld

THE CHALLENGE

The Fragmented Knowledge Landscape

Distributed Sources

Research isolated across faculty pages, center subdomains, and group portals.

Vocabulary Mismatch

Layperson terms fail against dense academic terminology like CGM or biosensing.

Zero Traceability

Standard AI search engines hallucinate unverifiable contacts and non-existent units.

OUR SOLUTION

The AI-Powered Research Navigator

1. **Unified Knowledge Graph**
   Aggregates people, groups, infrastructure, and active projects.

2. **Dynamic Lay-Term Expansion**
   Maps everyday queries to specialized research terminology.

3. **Verifiable Source Backing**
   100% of results link directly to original institutional URLs.

CORE PROPOSITION

"Connecting Bern's economic agency, industry partners, and public seekers directly with verified research capabilities."

Guaranteed Zero Hallucination

ECOSYSTEM SCOPE

One Unified Ecosystem

UniBE Subdomains

University of Bern

- Faculties & Institutes
- Research Centers & Labs
- Core Technology Platforms

ARTORG Center

Biomedical Engineering

- Artificial Organ Research
- Medical AI & Microfluidics
- Clinical Group Profiles

Diabetes Center

Diabetes Center Berne

- Translational Technology
- Metabolic Wearables
- Specialized Services

TECHNICAL ARCHITECTURE

End-to-End Pipeline Workflow

1. Web Crawler

Cleans HTML & parses structure.

2. Apertus 70B

Extracts fields & generates lay terms.

3. ChromaDB

Indexes vectors via MiniLM.

4. LLM Bouncer

Verifies facts & outputs JSON.

SEMANTIC INNOVATION

Dynamic Query Expansion

USER QUERY

"glucose sensors"

Common non-technical search phrase



EXPANDED ACADEMIC TERMS

CGM

continuous glucose monitoring

biosensing

metabolic monitoring

TRUST & ACCURACY

Strict LLM Match Verification

1. Vector Retrieval

Fetches top candidates using mathematical distance.

2. Fact Verification

Apertus 70B verifies candidates against extracted facts.

3. Match Reason

Generates 1 concise explanation sentence.

RICH DELIVERABLE

Structured Result Schema

Prof. Dr. Jane Doe

100% Traceable Match

Lead Researcher • ARTORG Diabetes Technology

FIELDS OF EXPERTISE

Continuous Glucose Monitoring, Biosensors, Wearables

RELATED PROJECTS

AI Glucose Sensor Platform, Closed-Loop Pump

Why matched: Expertise directly encompasses wearable biosensors and continuous metabolic monitoring.

VALIDATION

Success Criteria Fulfillment

80%+	100%	3 / 3	0
Top 10 Target Experts Found	Verifiable Source URLs	Institutional Ecosystems	Hallucinated Claims

FUTURE VISION

What Is Next: 4-Week Roadmap

WEEK 1

Auto Crawl Webhooks

Incremental background vector updates.

WEEK 2

Multilingual Alignment

Native German, French & English translation.

WEEK 3

Fine-Tuned Apertus 8B

Fully air-gapped on-premise execution.

WEEK 4

Network Graph UI

Visualizing interdisciplinary links.



Discover Bern's Innovation Engine.

Ready for deployment on Swiss Sovereign Cloud and air-gapped environments.

Team Apertus Bern